

HARISH S

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PROFESSIONAL SUMMARY

Driven **Machine Learning Engineer** with expertise in **Computer Vision**, **Generative AI**, and **RAG Architecture**. Hands-on experience developing end-to-end AI pipelines, fine-tuning LLMs, and deploying scalable web applications using **FastAPI**, **LangChain**, and **Streamlit**. Proven track record in transforming complex data into production-ready intelligence models.

TECHNICAL SKILLS

- **Programming & Core:** Python, SQL, REST APIs, Git, GitHub, Postman, VS Code
- **Machine Learning:** Scikit-Learn, Data Preprocessing, Feature Engineering, Model Evaluation, Joblib
- **Deep Learning & Vision:** Computer Vision, YOLOv8, OpenCV, Image Segmentation, Spatial Analysis
- **Generative AI & LLMs:** RAG Architecture, LangChain, FAISS, Sentence Transformers, Gemini 2.5 Flash
- **Deployment & Frameworks:** FastAPI, Streamlit, Microservices, Model Serving

FEATURED PROJECTS

AI Resume Analyzer using RAG Architecture | Python, LangChain, FAISS, Gemini 2.5 Flash, Streamlit

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) system to automate candidate evaluation and parsing.
- Integrated **FAISS** vector store and **Sentence Transformers** to perform semantic similarity searches across multi-page PDF resumes.
- Designed an interactive **Streamlit** user interface for natural language querying, optimizing profile retrieval workflows.

Earthwork Volume Estimation from Drone Videos | YOLOv8, OpenCV, Computer Vision, Python

- Architected a computer vision pipeline leveraging drone imagery to automate earthwork volume estimation and construction monitoring.
- Deployed **YOLOv8** object detection models for high-accuracy excavation/fill site identification under varying lighting conditions.
- Implemented spatial analysis algorithms and explored 3D terrain reconstruction techniques to improve measurement accuracy.

End-to-End Customer Churn Prediction System | Scikit-Learn, Feature Engineering, FastAPI, Streamlit, Joblib

- Developed a predictive machine learning model to identify high-risk churn customers using historical data.
- Executed robust feature engineering and hyperparameter tuning using **Scikit-Learn** to optimize model precision and recall.
- Built a web application in **Streamlit** with model persistence via **Joblib**, providing real-time risk scoring for end-users.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering

Sri Shakthi Institute of Engineering and Technology | **CGPA: 7.8 / 10**

CERTIFICATIONS & CONTINUOUS LEARNING

- **Oracle Agentic AI Course** – Oracle
- **Machine Learning Specialization** – DeepLearning.AI
- **Python Bootcamp** – Udemy